Human Health and Disease

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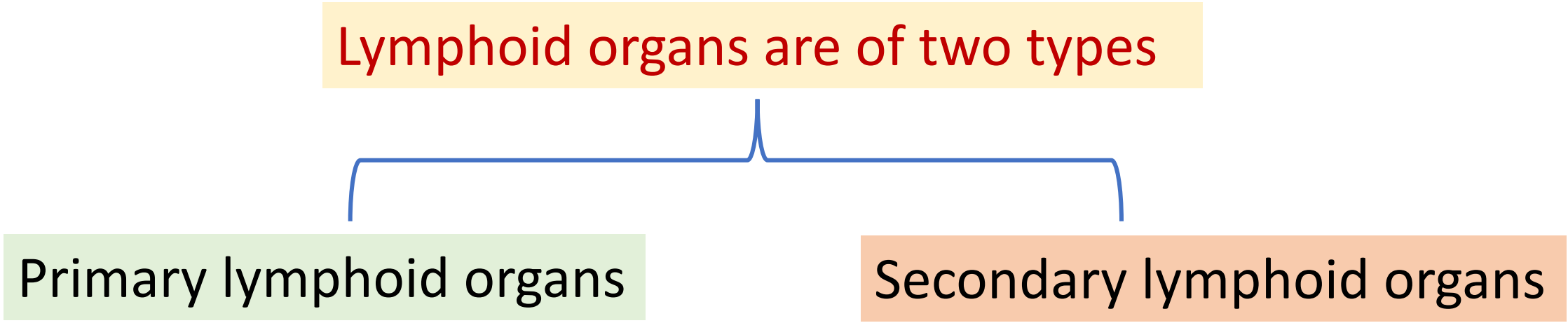
భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్

भारतीय प्रौद्योगिकी संस्थान हैदराबाद

Indian Institute of Technology Hyderabad

Lymphoid organs

These are the organs where origin, maturation and proliferation of lymphocytes occur.



Lymphocytes: Lymphocytes are a type of white blood cell. They play an important role in immune system, which helps the body fight disease and infection.

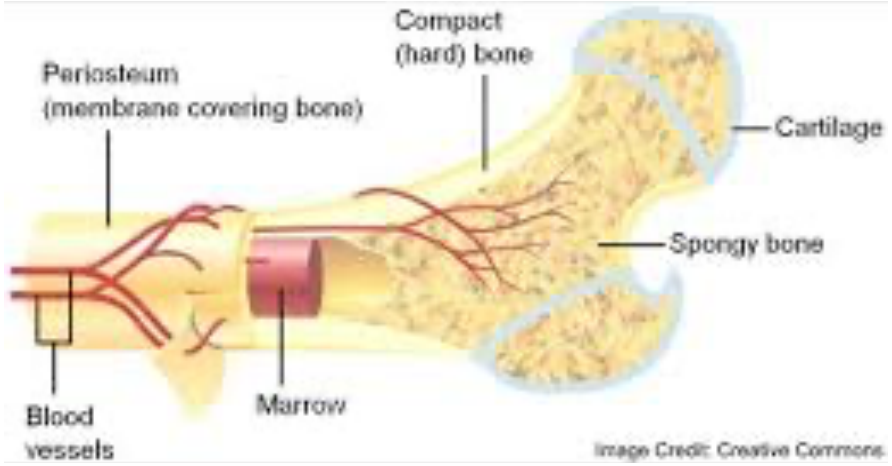
Immune system is made up of an intricate web of immune cells, lymph nodes, lymph tissue and lymphatic organs.

Primary lymphoid organs:

- The organs where immature lymphocytes differentiate into antigen-sensitive lymphocytes.
- 2 types of primary lymphoid organs: Bone marrow and Thymus

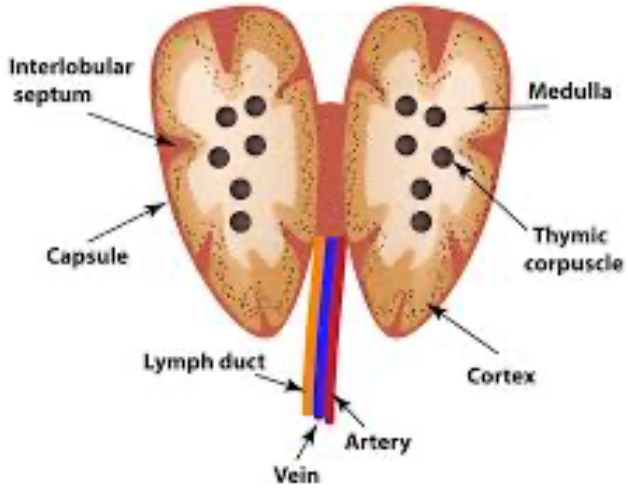
Lymphocytes: a type of immune cell (White blood cell)

Bone Marrow





Thymus



Primary lymphoid organs:

Bone Marrow:

- The cite of formation of all lymphocytes including B and T lymphocytes

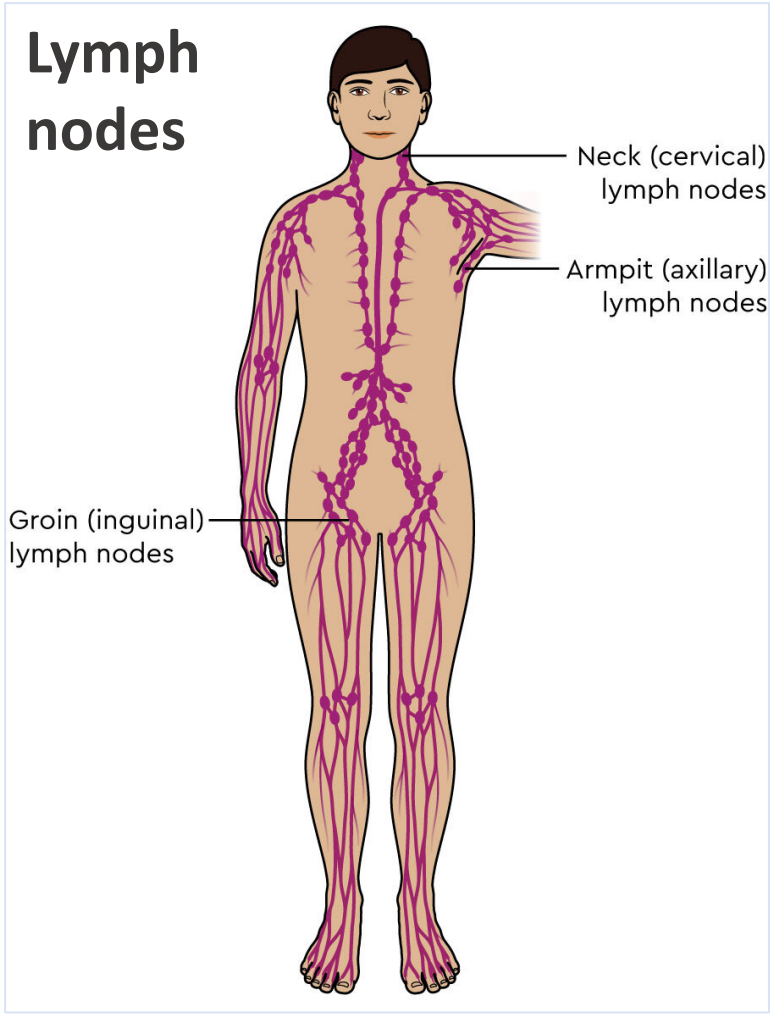
Thymus:

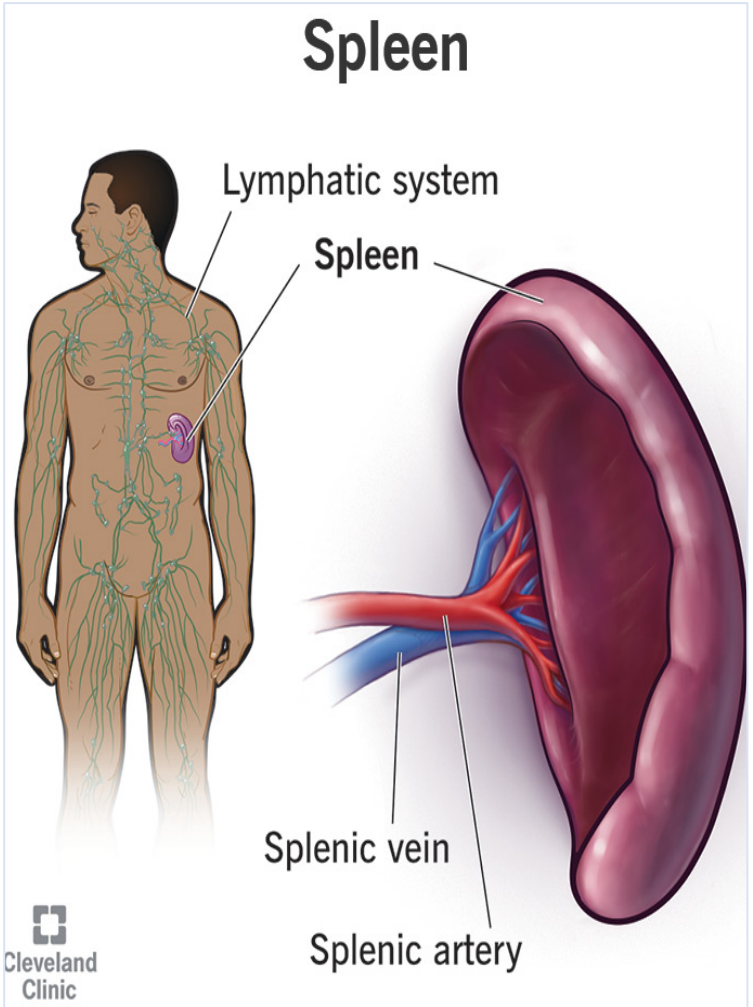
- A bilobed organ seen near the heart and beneath the breastbone.
- It is large during birth but gradually reduces in size and become very small size in puberty.
- Immature T-lymphocytes from bone marrow is migrated to thymus and matured.

Secondary lymphoid organs:

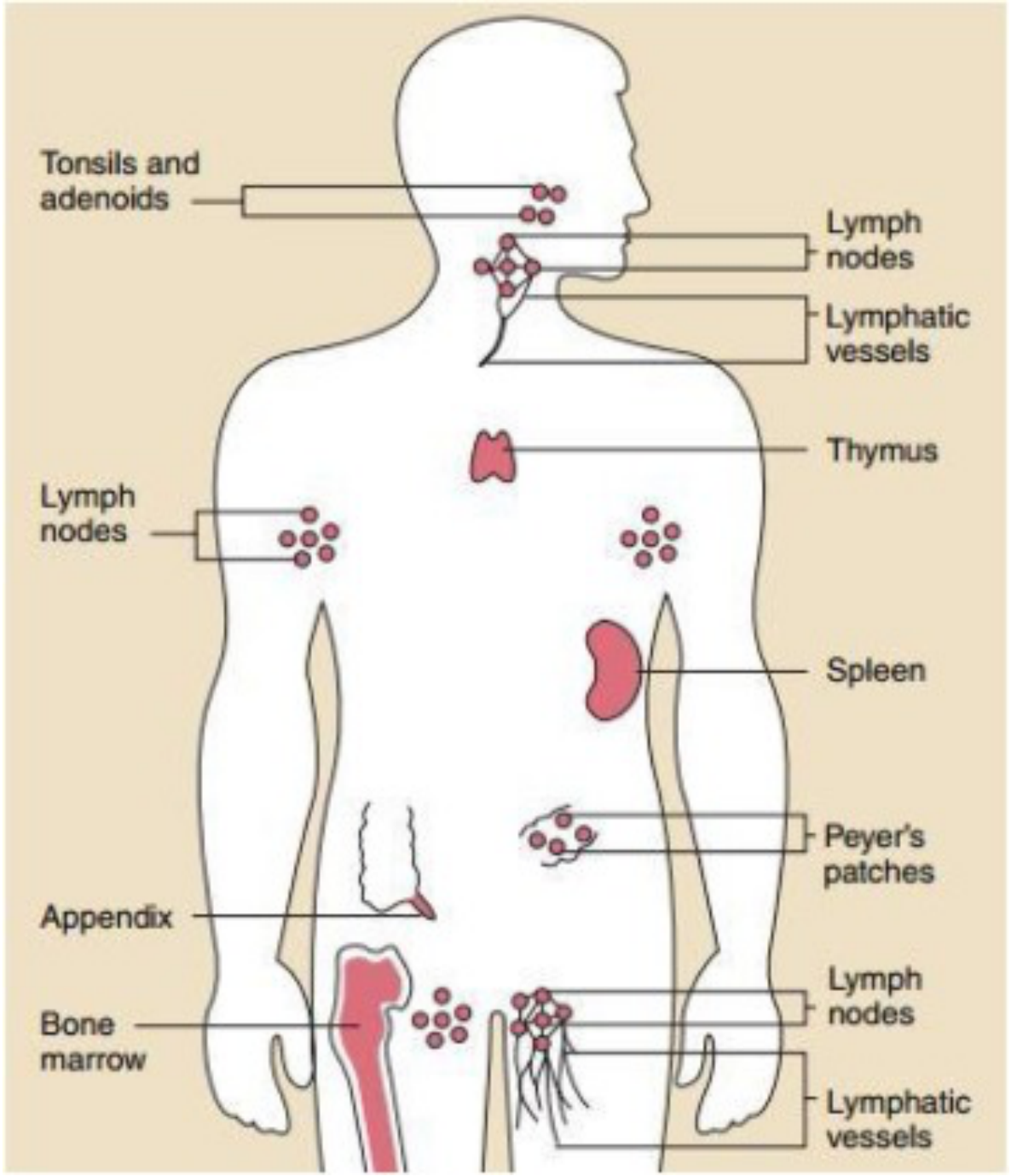
- The organs, to which matured lymphocytes migrate from primary lymphoid organs, interact with antigens and then proliferate to become effector cells

Examples: Spleen, Lymph nodes, tonsils, Peyer's Patches of small intestine and appendix





Location of various lymphoid organs in human body



Spleen:

- Bean-shaped organ.
- Contains lymphocytes and phagocytes.
- Removes worn-out RBCs and microorganisms from blood.
- Reservoir of erythrocytes in foetus

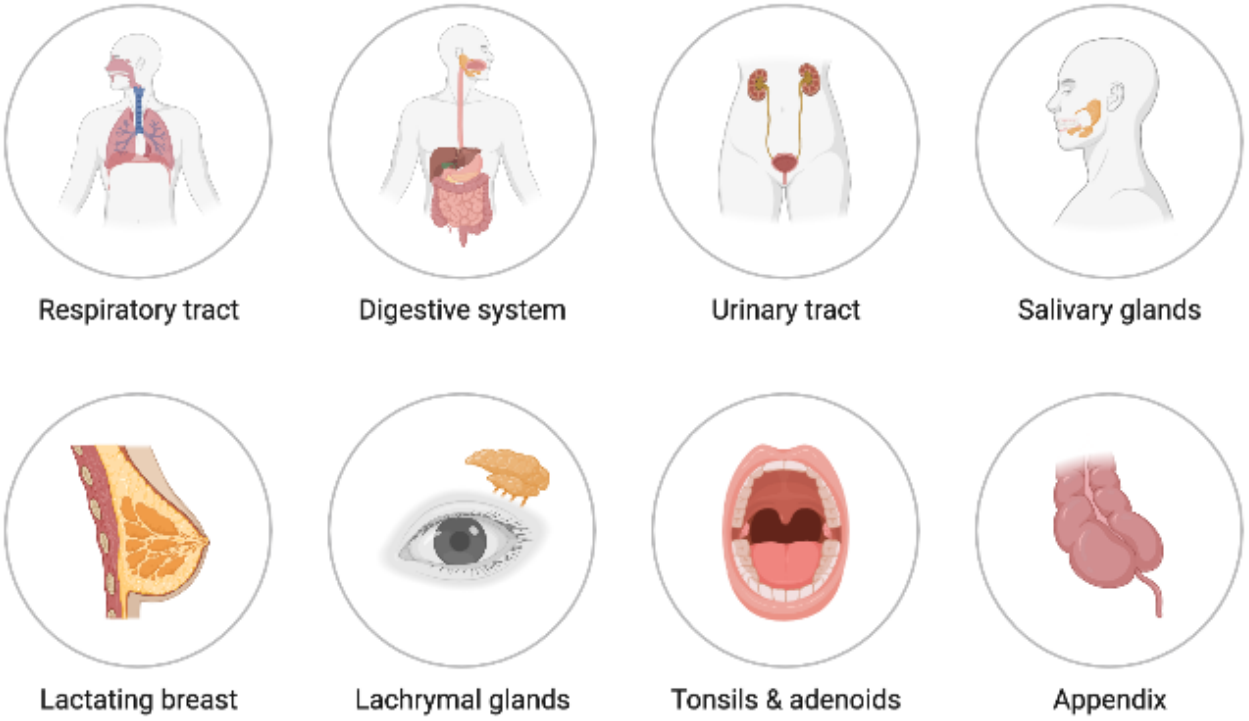
Lymph nodes:

- Found in lymphatic system.
- They trap microorganisms or other antigens.
- Trapped antigens activate lymphocytes and cause immune response.

Mucosa Associated Lymphoid Tissue (MALT)

- Is located within the lining of respiratory, digestive and urogenital tracts.
- It constitute 50% of lymphoid tissue.

Mucosal-associated Lymphoid Tissues
MALT



Immunity:

Ability of the host to fight the disease-causing organisms, conferred by the immune system OR resistance from disease based on prior exposure

- Immunity
  - Innate
  - Adaptive
    - Humoral
    - Cell mediated



Characteristic	Innate	Adaptive
Response time	Quick/immediate	Delayed
Cellular and chemical barriers	Skin, mucosal epithelia; antimicrobial molecules	Lymphocytes in epithelia; antibodies secreted at epithelial surface
Secreted proteins	Complement, various lectins	antibodies
Cells	Phagocyte (macrophages, neutrophils), dendritic cells, natural killer cells, mast cells, innate lymphoid cells	Lymphocytes (B and T cells)
Response to repeat infection	Same each time	More rapid and effective with each subsequent exposure

Characteristics	Innate	Adaptive
Specificity	For molecules shared by groups of related microbes and molecules produced by damaged host cells (non-specific)	For many different microbial and non-microbial antigens (Specific)
Diversity	Low; recognition molecules encoded by inherited (germline) genes *Ex: Complement system, PRRs, physical and chemical barriers*	Very high; many antigen receptors are generated by somatic recombination of gene segments in lymphocytes
Memory	Limited or No	Yes

Adaptive immunity is of two types:

1: Humoral immunity

('Humors' meaning fluid)

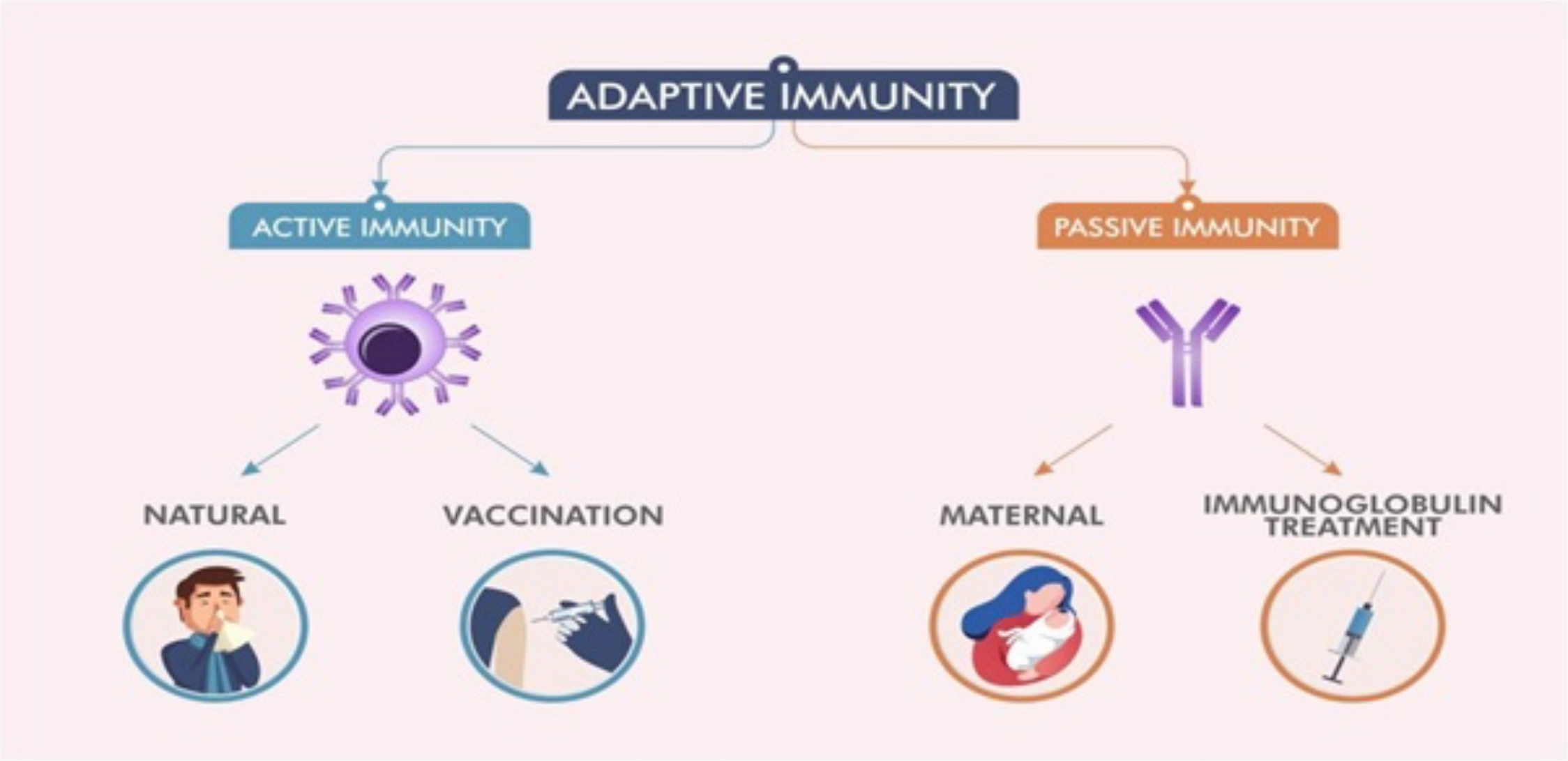
- Protein mediated i.e antibodies
- Types of cells which secretes antibodies is B-cell
- Induce by *extracellular* microbe

2: Cell mediated immunity

- Cytokines mediated
- T cell mediated

- Induce by phagocytosed microbe which reside in macrophage and by *intracellular* microbes which replicate inside the cell

Adaptive immunity is also divided into:



Pathogens Comes in Many Forms and Must First Breach Natural Barriers

Pathogens: Organisms which causes disease
Pathogenesis: Process by which pathogen induces diseases

Pathogen and Immune system interactions depends on the

- Nature of the invader
- Microenvironment in which the engagement happens

Immune system do not get engage until the pathogens breach the natural physical barriers:

- Skin
- pH (Acidity of stomach, Vagina, Persipiration)
- Mucous membrane

Types of barriers in innate immunity

4 types of barriers:

1. Physical barriers:

Skin is the main barrier which prevents entry of the micro-organisms

Mucus coating of the epithelium lining the respiratory, gastrointestinal and urogenital tracts helps in the trapping microbes entering the body

2. Physiological barriers:

Acid in stomach, Saliva in the mouth, tears from eyes-all prevents microbial growth.

3. Cellular barriers:

Leukocytes (White blood cell (WBC), like neutrophils, monocytes (in blood), macrophage (in tissue), natural killer cell phagocytes and destroy microbes.

4. Cytokine barriers:

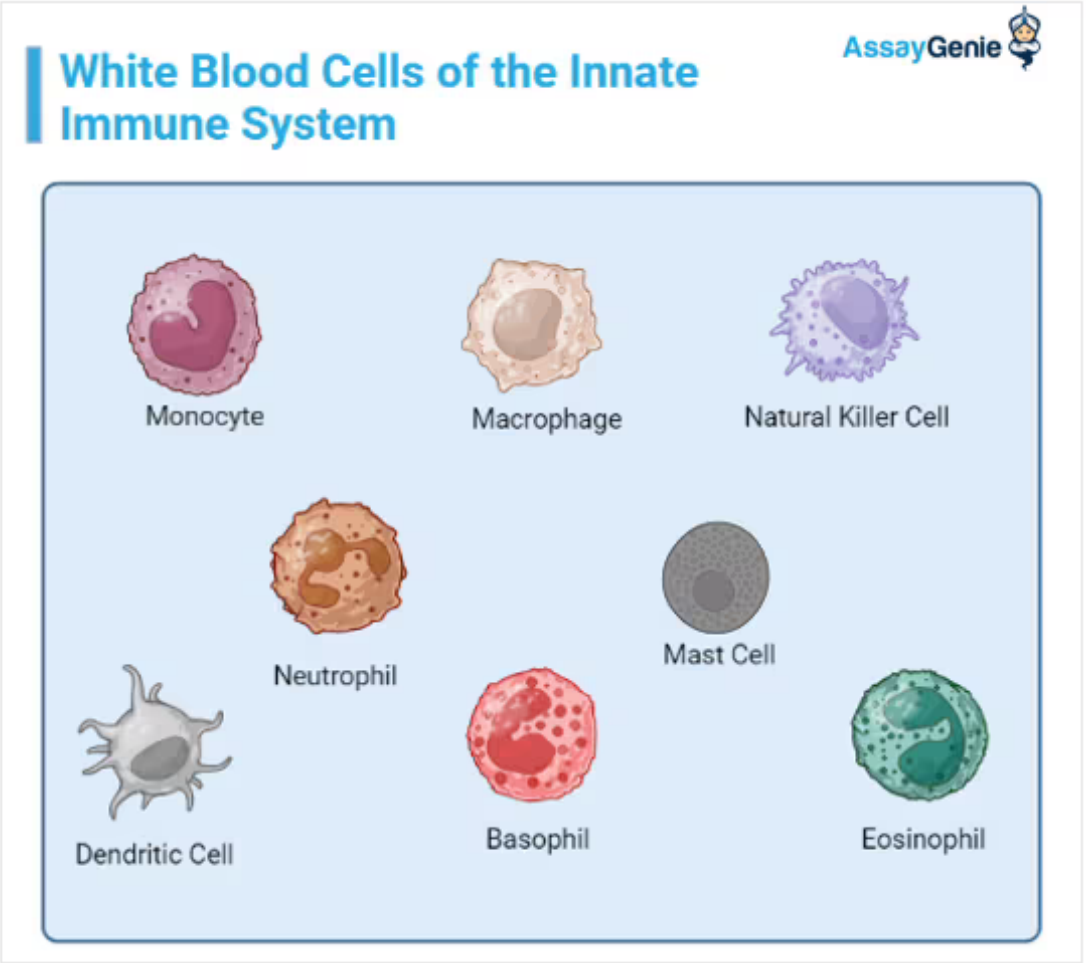
Virus-infected cells secrete proteins called **interferons** which protect non-infected cells from further viral infection.

Cells Involved In Innate Immunity

Phagocytes: These circulate through the body and look for any foreign substance. They engulf and destroy it defending the body against that pathogen.

Macrophages: These have the ability to move across the walls of the circulatory system. They release certain signals as cytokines to recruit other cells at the site of infections.

Mast Cells: These are important for healing wounds and defense against infections.

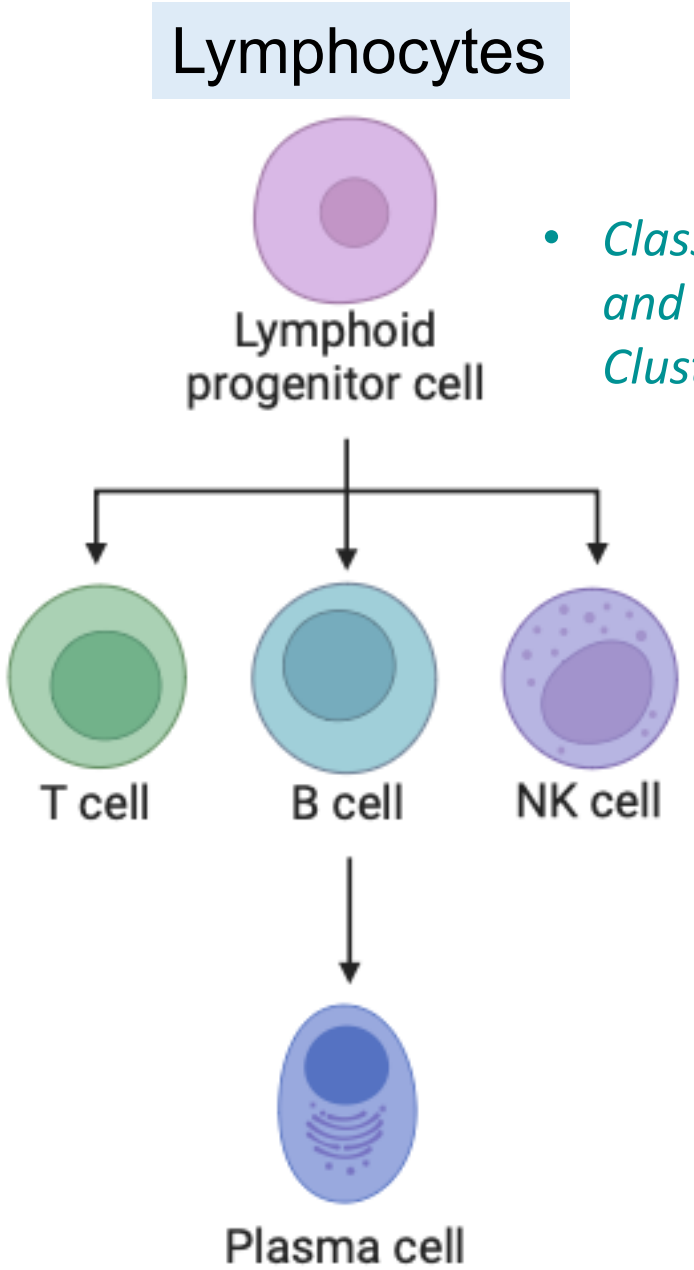


Neutrophils: These contain granules that are toxic in nature and kill any pathogen that comes in contact.
Eosinophils: These contain highly toxic proteins that kill any bacteria or parasite in contact.
Basophils: These attack multicellular parasites. Like the mast cells, these release histamine.
Natural Killer Cells: These stop the spread of infections by destroying the infected host cells.
Dendritic Cells: These are located in the tissues that are the points for initial infections. These cells sense the infection and send the message to the rest of the immune system by antigen presentation.

Cells of the Lymphoid Lineage Regulate the Adaptive Immune Response:

Lymphocytes:

- Principal cell players in the adaptive immune response
- Represents 20-40 % of circulating WBC and 99% of cells in the lymph



- *Classified based on functional and phenotypic differences [Ex: Cluster of Differentiation (CD)]*

Cluster of Differentiation (CD) or cluster of designation

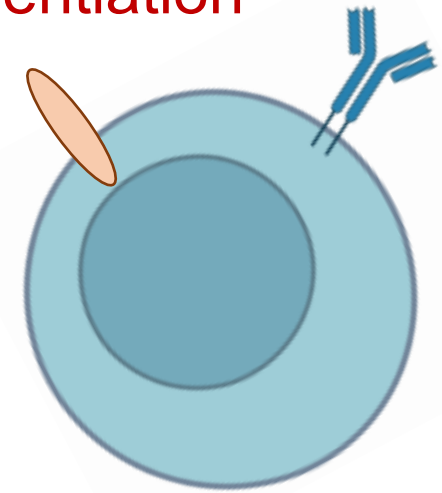
- B and T cells are differentiated phenotypically by surface Cluster of Differentiation (CD), often called “markers”
- These are the Surface proteins (antigens) expressed by immune cells
- As cells mature, they express different protein receptors on the cell surface, which can aid in determining the type and maturation stage of the cells being examined
- In addition to their CD surface signatures, each B or T cell also expresses an antigen-specific receptor (the **B cell receptor (BCR)** or the **T cell receptor (TCR)**, respectively) on its surface.

B cell

Cluster of Differentiation (CD)

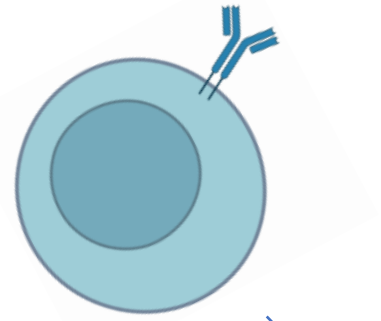
B cell Receptor (BCR) or Immunoglobin or antibody

Membrane bound antibody that binds to antigen



- Mature in bone marrow (*bursa* of Fabricius of birds) meaning bone marrow
- Mature B cells distinguish themselves from other lymphocytes by the expression/display of BCR

Activated B cell



Differentiate into effector cell known as

Memory Cell



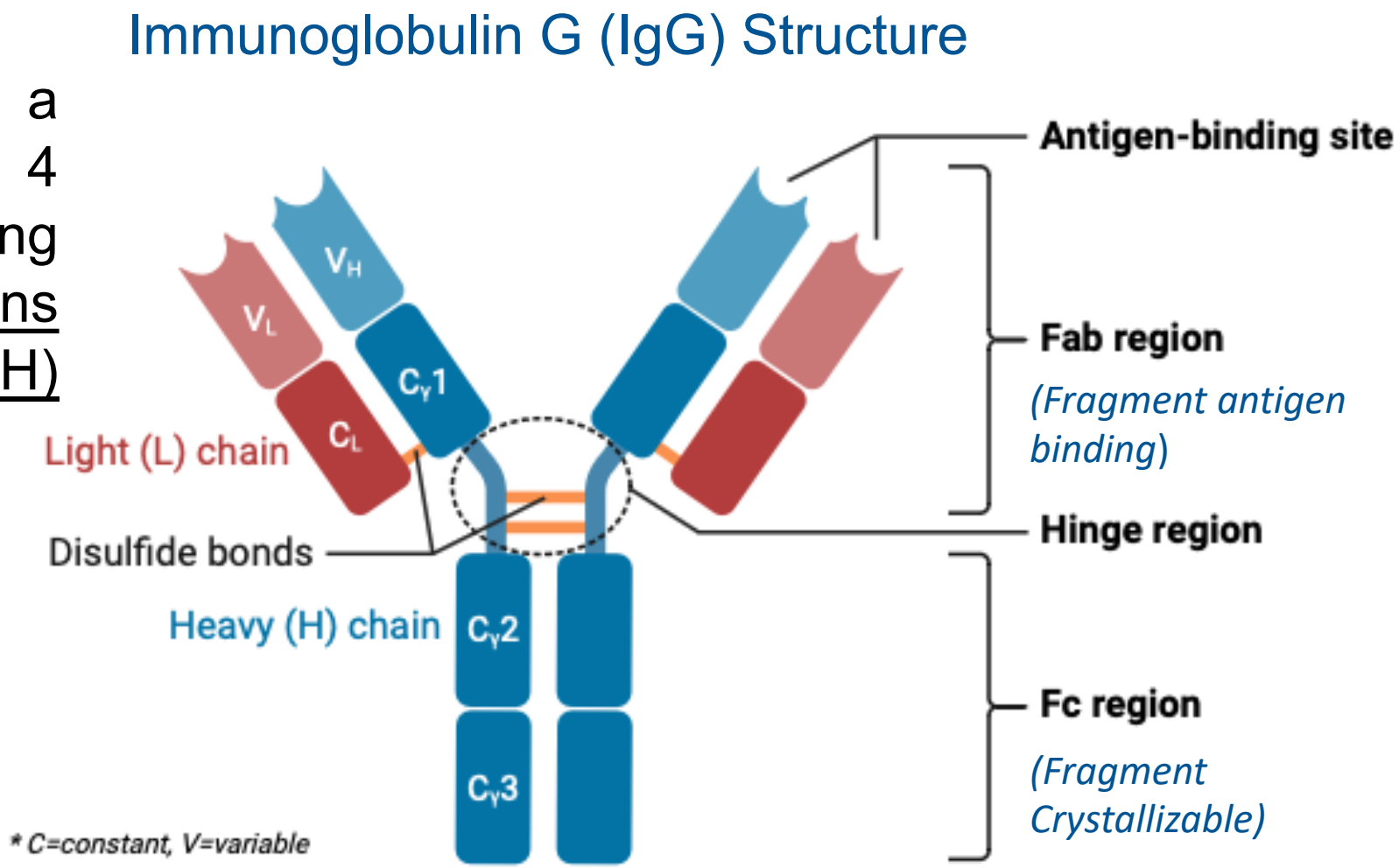
Plasma cell

- Do not divide
- Secretes antibody

The Structure of Antibodies/Immunoglobulins

- All antibodies (Ab) share a common structure of 4 polypeptide chains, consisting of two identical light (L) chains and two identical heavy (H) chains

- Disulfide bonds: Heavy and light chains are connected to each other *via* Disulfide bonds



Antibody/Immunoglobulin function

- Antibodies protect the host against infection, by binding to pathogens & facilitating their elimination
- Immunoglobulin constitute the 20 -25 % of total serum proteins

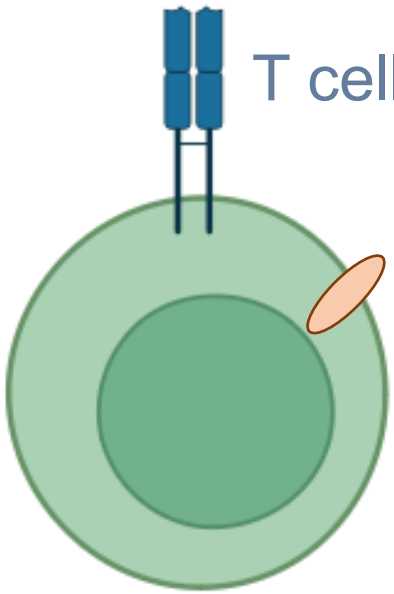
Classification of Antibody/Immunoglobulin:

Antibody Class	Types of heavy chain	Types of light chain
IgA	Alpha (α)	κ (kappa) or λ (lamda)
IgD	Delta (Δ)	κ or λ
IgE	Epsilon (ϵ)	κ or λ
IgG	Gamma (Y)	κ or λ
IgM	Mu (μ)	κ or λ

Type	Main characters and Occurrence	Functions
IgA	• Present on mucus membrane, gut, urogenital tract and colostrum (first milk after giving birth) • Secreted into mucous, saliva, tears, colostrum	• Protection of mucous membrane and outer surface of the body • Protection from inhaled and ingested pathogen
IgD	• Present on the naïve (exposed) B cells • Activate mast cells and basophil	• Activation of B cells • Development and maturation of immune reactions
IgE	• Present in very small quantity • Binds to allergens • Involved in tackling parasitic infections	• Stimulation of mast cells, related to allergic reaction • Protection from Parasites

Type	Main characters and Occurrence	Functions
IgG	• Most abundant, Main immunoglobin of blood and intestinal fluid • Can cross the placental barrier and provide protection to the fetus • Lightest antibody	• Stimulate complementary system • Provide protection to embryo • Role in phagocytosis
IgM	• First antibody secreted in response to antigens • Heaviest antibody • Pentameric form	• Strong agglutination • Related to complement system

T cell



- Mature in **Thymus**, derived their letter designation from the site of maturation
- Do not secrete antibody, but help B-cell to produce them
- Express a unique antigen-binding receptor called the T-cell receptor (TCR)
- TCR only recognize processed pieces of antigen (typically peptides) bound to cell membrane proteins called **major histocompatibility complex (MHC)** molecules

Naïve vs. Immune Individuals

